

SENTIMENT ANALYSIS OF STUDENT FEEDBACK USING BERT

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ABSTRACT

This study applies BERT for sentiment analysis of student course evaluations at CvSU, achieving 91.7% accuracy and enabling automated identification of positive and negative instructional feedback patterns.

INTRODUCTION

Student course evaluations generate large volumes of unstructured text that are often analyzed manually, introducing bias and delay. Natural Language Processing (NLP) can automate sentiment classification. This study fine-tunes BERT on a dataset of 8,500 student feedback entries to classify sentiments as Positive, Neutral, or Negative, supporting data-driven faculty performance evaluation.

METHODOLOGY

BERT-base-uncased was fine-tuned using the HuggingFace Transformers library on a dataset of 8,500 course evaluation comments collected from CvSU's Student Evaluation System (SES). Labels were assigned by two human annotators with inter-rater agreement of Cohen's Kappa = 0.83. A stratified 70/15/15 train/validation/test split was used. Hyperparameters: batch size 32, learning rate 2e-5, 5 epochs, max token length 128.

RESULTS

BERT achieved 91.7% accuracy on the test set. Macro F1 was 0.90. Positive class precision was 94%, Neutral 87%, and Negative 93%. BERT outperformed baseline logistic regression (78%) and BiLSTM (85%). The model also produced interpretable attention maps highlighting key sentiment-bearing phrases.

DISCUSSION

BERT's contextual embeddings significantly improve feedback sentiment classification over traditional ML methods. Neutral class confusion with Positive remains a challenge due to polite but critical feedback patterns common in Filipino academic culture. Future integration with the university's evaluation portal would enable real-time sentiment dashboards.